

# Analysis of Delay and Doppler Spreads in Measured sub-6 GHz and mmWave Time-Varying Channels

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**Abstract**—One of the key research directions to increase the capacity of new radio (NR) vehicle-to-everything (V2X) communication systems is the extension of employed frequency bands from sub-6 GHz to the millimeter wave (mmWave) range. To investigate different propagation effects between sub-6 GHz and mmWave bands in high-mobility scenarios, one needs to conduct channel measurements in both frequency bands. Using a suitable testbed setup to compare these two bands in a fair manner, we perform channel measurements at center frequencies of 2.55 GHz and 25.5 GHz, velocities of 50 km/h and 100 km/h, and at 126 different spatial positions. Furthermore, we conduct a comparative study of the multi-band propagation based on measurement results. We estimate the power delay profile (PDP) and the Doppler power spectral density (DSD) from a large set of measurements collected in a measurement campaign. Finally, we compare measured wireless channels at the two employed frequency bands in terms of root-mean-square (RMS) delay spread and RMS Doppler spread.

**Index Terms**—mmWave, sub 6 GHz, 5G, statistical evaluation, vehicular communications, testbed, channel measurements, RMS delay spread, RMS Doppler spread.

## I. INTRODUCTION

Future vehicular applications and services require high data transmission rates that cannot be achieved in conventional sub-6 GHz bands. A possible answer to this growing demand for high transmission rates can be found in millimeter wave (mmWave) bands. To learn how propagation and scattering effects change from sub-6 GHz to mmWave bands, comparative measurements over different bands have to be conducted. There are few comparative multi-band analyses in the literature considering static [1] and drive-by vehicular scenarios [2].

Static measurements allow the analysis of the frequency selectivity but not the time selectivity of a vehicular channel. On the other hand, drive-by measurements allow for both time- and frequency-selectivity analysis but pose the problem of antenna placement. Due to mechanical reasons, different antennas for different frequency bands cannot be placed at the same position at the same time. Placing antennas for different frequency bands several centimeters apart leads to a position offset of several wavelengths at mmWave frequencies. This difference in antenna position may result in a different small-scale fading behavior or altered channel statistics. Therefore, one needs to perform high-mobility wireless channel measurements with identical transmit and receive antenna positions.

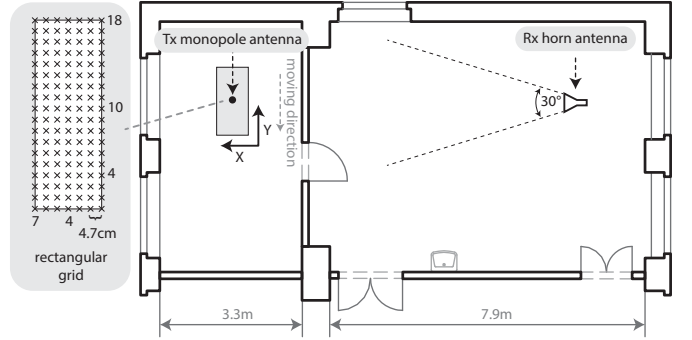


Fig. 1. Measured indoor laboratory environment. The moving transmit antenna and the static receive antenna are located in neighboring rooms.

## II. MEASUREMENT CAMPAIGN

Our testbed setup allows repeatable measurements at sub-6 GHz and mmWave bands in a controlled environment at high velocities with identical transmit and receive antenna positions. The setup is described in [3] and consists of a moving transmitter and a static receiver. The moving transmitter is based on a rotary unit which rotates an antenna around a central axis at a constant but adjustable velocity. The whole rotary unit is placed on a sliding board, which can be moved by 33 centimeters along the x-axis and 81 centimeters along the y-axis.

Using the testbed setup, we perform channel-sounding measurements in an indoor environment as shown in Fig. 1. We conduct measurements with specific center frequencies and velocities in scenarios A (2.55 GHz and 50 km/h), B (2.55 GHz and 100 km/h), C (25.5 GHz and 50 km/h) and D (25.5 GHz and 100 km/h) at different spatial positions according to the rectangular grid shown in Fig. 1. As a result of the measurement campaign, we obtain a set of time-variant channel transfer function (CTF) for discrete-time (snapshots) and frequency (subcarriers).

To analyze the time-frequency dispersion, we characterize the channel by the local scattering function (LSF) using the concept from [4]. For each measured scenario, we assume the channel to be locally stationary within a window of approximately  $6\lambda$  of motion and over the entire frequency range, without further justification. A window of  $6\lambda$  corresponds

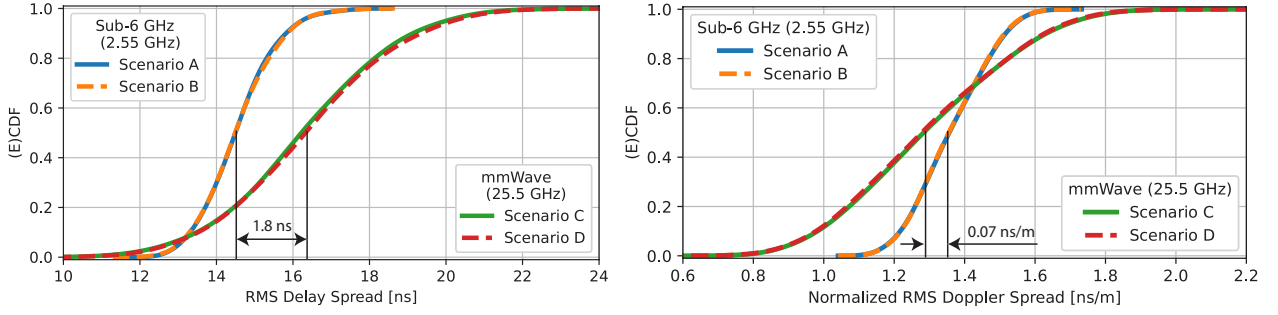


Fig. 2. RMS Delay Spread and normalized RMS Doppler Spread are almost invariant with regard to the transmitter velocity.

to  $L = 250$  snapshots for the 2.55 GHz case and  $L = 25$  snapshots for the 25.5 GHz case. The LSFs of each window are then averaged to produce an average LSFs.

After that, we calculate the power delay profile (PDP) and Doppler power spectral density (DSD) as the expectation of the average LSF over the Doppler domain or the delay domain, respectively. To allow for better comparability between different frequency bands and different velocities, we perform the following adjustments. Firstly, we normalize the power of the PDP and DSD such that 0 dB is the greatest occurring value. Secondly, we define the dynamic range (DR) as the difference between maximum power and noise floor and consider the values outside the DR as a corresponding noise floor. Thirdly, we define the normalized Doppler shift such that the DSD is confined to the same interval for all measured scenarios and does not depend on a particular center frequency and velocity. Finally, we calculate the root-mean-square (RMS) delay spread and normalized RMS Doppler spread as second-order moments of the PDP and normalized DSD, respectively. The RMS delay and normalized RMS Doppler spreads are directly related to the coherence bandwidth and coherence time of the channel, which indicate the rate of change of the channel in frequency and time, respectively.

### III. RESULTS AND DISCUSSION

We perform the statistical evaluation of the channel spreads by calculating RMS spreads for each position according to the rectangular grid from Fig. 1. We plot in Fig. 2 the cumulative distribution function (CDF) of the RMS spreads for all stationarity regions and spatial positions.

One can observe a higher standard deviation for the mmWave scenarios compared to the sub-6 GHz scenarios. This occurs due to the following phenomena. Firstly, for 25.5 GHz the stationarity region of  $6\lambda$  is spatially smaller. By performing measurements over the same path (arc segment from  $-40^\circ$  to  $40^\circ$ ), we obtain more independent realizations of mmWave channels than sub-6 GHz channels. Secondly, the distance of 4.7 cm between different spatial positions (see Fig. 1) corresponds to  $0.4\lambda$  for the sub-6 GHz scenarios and  $4\lambda$  for the mmWave scenarios.

Furthermore, it can be observed that for the mmWave band, the mean RMS delay spread is larger by about 1.8 ns (12%).

However, this difference is negligible on a large scale and should not significantly impact the system design. The mean values of the normalized RMS Doppler spread are similar for different frequency bands as well, with the fact that the sub-6 GHz scenarios have a slightly higher value by about 0.07 ns/m (5%) compared to the mmWave scenarios. Due to the spatial filtering by the receive antennas, the mean values are 40% and 38% less than the maximum normalized RMS Doppler shift ( $\nu_{N,\max} = 3.33$  ns/m) for the sub-6 GHz and mmWave scenarios, respectively.

One can notice that both spreads are almost invariant with regard to the transmitter's velocity, although the measurements were performed at different times. The largest mean difference between the low-velocity and high-velocity scenarios within the same frequency band is around 0.7%. Thereby, we confirm the accuracy of our testbed proposed in [3], since such a deviation between the low-velocity and high-velocity scenarios is negligible on a large scale.

### IV. CONCLUSION

Our testbed to compare different frequency bands in a fair manner works correctly and is able to reproduce an experiment with minor uncertainties. The measured results show small differences on the RMS delay spread and normalized RMS Doppler spread, thereby offering almost the same propagation conditions at different frequency bands. The antenna directivity impacts the RMS Doppler spread to a great extent.

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